SCAR: Smart Cart based on ARM algorithm and RFID technology

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***Abstract*—A supermarket is a place where people get all products in one place. Where supermarkets are more advantageous for shopping, there are also few disadvantages that make people leave supermarkets at billing counters due to long queues at checkout counters for billing. The most irritating part during the shopping in supermarkets is waiting in the long queues for billing at checkout counters and having lack of product information, which wastes more time at checkout counters. The proposed system SCAR will give the customers a quick shopping experience by facilitating quick billing system attached to cart using RFID technology and a friendly product recommendation on their smartphones using ARM algorithm. Customer will be aware of shopping is under budget or over budget. Product recommendation sent directly to a smartphone will arouse customer's purchasing desire. Also, this system will lessen the billing staff requirement and will increase the sales productivity. The hardware required to implement this system is inexpensive.**

Keywords—Product Recommendation, Association Rule Mining, Eclat, RFID Technology

I. INTRODUCTION

The supermarket's culture turned out to be a big retail business, as they provide various types of products. Visiting a grocery mall is advantageous because these supermarkets or retail systems provide us all daily necessities ranging from groceries, food products, electrical appliances, clothing, foot wares, etc. all in one place. Most of the time, customers face many problems while shopping, such as incomplete information about new products or trendy products, worrying that amount of money carried is sufficient or not, and also, waiting for a long time in queues for billing at checkout counters is a tedious task while shopping. Improvement is required in the existing system to make shopping easier and friendly for customers and also beneficial to supermarket owners.

Similar to the online e-commerce system, offline systems such as supermarkets or retail systems needs product recommendation system, to increase their sales. This can be achieved by identifying frequently bought together products by generating frequent itemsets and rules using an association algorithm. This information is used for product recommendation or sales marketing for visiting customers [1], [2].

RFID is listed in the most important automatic identification technologies. These technologies include Barcode systems, QR code systems, Biometric procedure, Smart cards, and RFID systems. This type of technology collects identification data from objects. The collected object's data is further used to uniquely identify these objects and keep a record of them. These technologies are more accurate and faster as compared to human manual work. In many ways, RFID technology exists around us today, even though they are habitually obscure to consumers. Its capability to keep the record of moving objects fostered the RFID technology.

RFID technology offers a flexible system for shopping malls, which improves the way of shopping for the customers in any self-service supermarkets or retail store. As the new, inexpensive and small-sized paper and sticker RFID tag developing techniques have emerged, RFID technology is increasingly encouraging to replace the barcode system [3]. Table 1 shows the Comparison of Barcode technology with the RFID technology.

TABLE I. COMPARISON OF BARCODE AND RFID TECHNOLOGY.

Attribute	**Barcode**	**RFID**
Accessibility Range	⩽ few inches	⩽ few meters
Line of Sight	Required	Not Required
Read/Write	Read Only	Read and Write
Information Capacity	Very Less	More than Barcode
Reliability	Damaged tags won't work	Damaged tags work perfectly
Cost (in bulk)	$0.001	$0.01

To overcome the above problems of long queues and product recommendation in offline retail supermarkets, we propose a system which is based on RFID technology and ARM algorithms.

II. RELATED WORK

A. Automatic billing system

The purpose of [4] is to provide a feature of an automatic billing system which is implemented using two technologies namely, RFID and ZigBee technology. This system decreases the waiting time in queue for billing at the checkout counters of malls or supermarkets.

As the Zigbee technology is having a range of 0 to 30 meters and Wi-Fi technology is having a range of 30 to 300 meters, Wi-Fi technology is more preferable in this SCAR system for wireless communication. This will decrease the number of access points required for server and cart communication in supermarkets.

B. Product Recommendations

Based on shopping trajectories and Wi-Fi logs [5]: The author of this paper comprises the data collection by using Wi-Fi logs and shopping trajectories in a non- intrusive way using LCCF, TA-LCCF, LVM and TA-LVM (proposed) methods. The methodologies used in this paper for recommending products are not suitable for the supermarkets or grocery shops.

Based on Behavior of Customer [6]: The author of this paper comprises the data collection by sensing values from the shopping cart and taking appropriate decisions by the server. This system helps the customer to select the products and thus arouses customer's purchasing desire. To allow the customers to easily use this system, the author of [6] adopted two ordinary actions or behaviors of a customer regarding the handling of the cart that observes his/her behavior: (a) Holding the handle of the cart, and (b) motion of the cart. These two actions are combinedly observed and used by the author of [6] to infer the nine possible cases which will be used to observe the customer behavior. Those behavioral results include having no, moderate or significant interest in the product. Also, whether the customer requires any assistance or not, depending on the number of times he visits the same shelf. In addition, to save the energy of the cart's battery, the cart's system is put into standby mode, depending on the time the cart is idle or in use. The limitation of this system is that it requires to install Zigbee routers at each rack to know the customer's location and their interest in type of products

Based on Search Keywords[7]: Author of paper [7] had used relevance between specific search keywords to specific products to know the preference of the user. The author first performed mining on historical user browsing data to build the connection between searched keywords and products to be recommended. These connections are represented in a graph form. Then relevance between search keywords and products is derived from points of the graph. when a new user starts any search, his/her search keywords are used for knowing his preference and then related products are recommended to the customers. Some times keywords may have different meanings in different contexts which may result in inappropriate recommendations.

III. PROPOSED SYSTEM AND METHODOLOGY

The proposed system Smart Cart based on ARM algorithm and RFID technology, denoted as SCAR, is the combination of the quick billing system and product recommendation system. The product recommendation system is completely dependent on the quick billing system, as the product recommender systems need to know the type of products present in the customer's cart.

A. Quick Billing System (QBS)

In Quick billing systems, RFID technology is used to solve the problem of waiting for a long time in the queue for billing at the checkout counters of supermarkets and saves the time of the customer as well as the cashier. The quick billing system updates the total bill of the cart on the display screen attached to the cart at each scan of the product. For adding the product to the cart, the customer needs to scan the product and drop into the cart. While, for removing the product from the cart, the customer needs to press remove button and then scan the product within 10 seconds. RFID tags and reader are used to implement this QBS system.

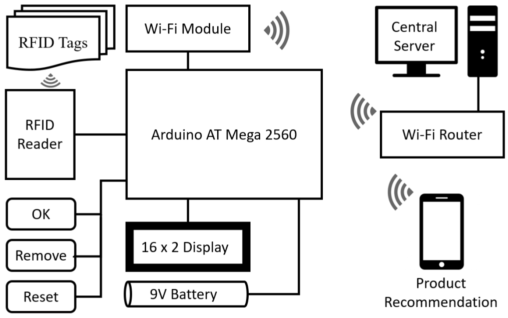

Fig. 1. Block Diagram of SCAR System.

The proposed system assumes that each product comes with an RFID sticker or paper tag attached to it from manufacturing factory. In this SCAR system, each cart is attached with an electronic box containing Arduino ATmega2560, ESP8266 Wi-Fi module, 16x2 LCD Display, MIRF522 RFID Reader, MIRF522 RFID Tags attached to the product, TTP-223 Touch keys, Wire, Battery, and the capacitor. If a customer wants to quit the shopping in-between the process, he/she can press the RESET button and then the OK button for confirming reset. The Billing process is managed centrally by the server and after checkout, the server resets the cart. Fig. 1 shows the block diagram of the proposed system.

B. Product Recommendation for proposed system

Product recommendation involves displaying of top selling products in real time, based on the products already dropped into the cart, and frequently bought products by old customers.

Steps to be taken at Customer-Side: Each cart will be provided with a QR code sticker which will be unique. The customer has to connect his/her smartphone to mall's Wi-Fi access point and then needs to scan the QR code by any QR code scanner Application. After scanning the QR code, a URL with the unique cart code is displayed on the smartphone. The customer needs to click on the link to open the required web page. Then he/she can see top selling products, recommended products similar to products present in the cart, and top sold products on his/her smartphone.

Steps to be taken at Server-Side on customer checkout: At each checkout of the customer, the server runs the ARM algorithm i.e. Eclat Algorithm as given in [8], [9] and [10], to generate frequent itemsets and rules using transaction history. This algorithm helps the server for recommending the products to the customers present in the mall using frequently bought products by previous customers. These frequent products are further categorized based on similar products and top sold products.

Steps to be taken at Server-Side on page load/refresh:

- Run CPR algorithm.
- Retrieve Product IDs from the rules generated by Eclat algorithm.
- Display the products with the same category of products present in the cart.

- Display top sold products those are bought frequently by previous customers.

Algorithm 1 CPR (CartProductRecommender)

Input: CSC, APC, T
Output: PL
Function:
  TL $\leftarrow$ APC $-$ CSC
  for each k in TL **do**
    $P_k \leftarrow [\text{count}(p_k) * 100] \div C_{total}$
  end for
  j $\leftarrow$ 0
  for each i in TL **do**
    if $\text{TL}_i \geq T$, **then**
      $\text{PR}_j = \text{TL}_i$
      j++
    end if
  end for
  for each i in PR **do**
    display PR_i
  end for

(Following are the definition of variables used in the algorithm:
  CSC: List of products existing in a customer's self-cart,
  APC: List of all products existing in all carts,
  TL: Temporary list of products,
  PR: List of products to be recommended,
  P: Percentage of products over total carts,
  T: Threshold,
  C: Cart,
  p: product id**)**

IV. System Details

Fig. 2 shows the Internal Hardware configuration and outer view of the SCAR system. Fig. 3 shows the conceptual view of SCAR system when attached to the cart and the conceptual view of communication between the cart and the server through Wi-Fi router. A server is connected to a Wi-Fi router. All the carts and smartphones are connected to this Wi-Fi access point. Fig.4 shows the flow of SCAR system. This flow shows the working process of both customer side and server side.

Internal Hardware Setup   Customer Facing Display

Fig. 2. Hardware Configuration.

(a)   (b)

Fig. 3. Conceptual view of SCAR system and network communication.

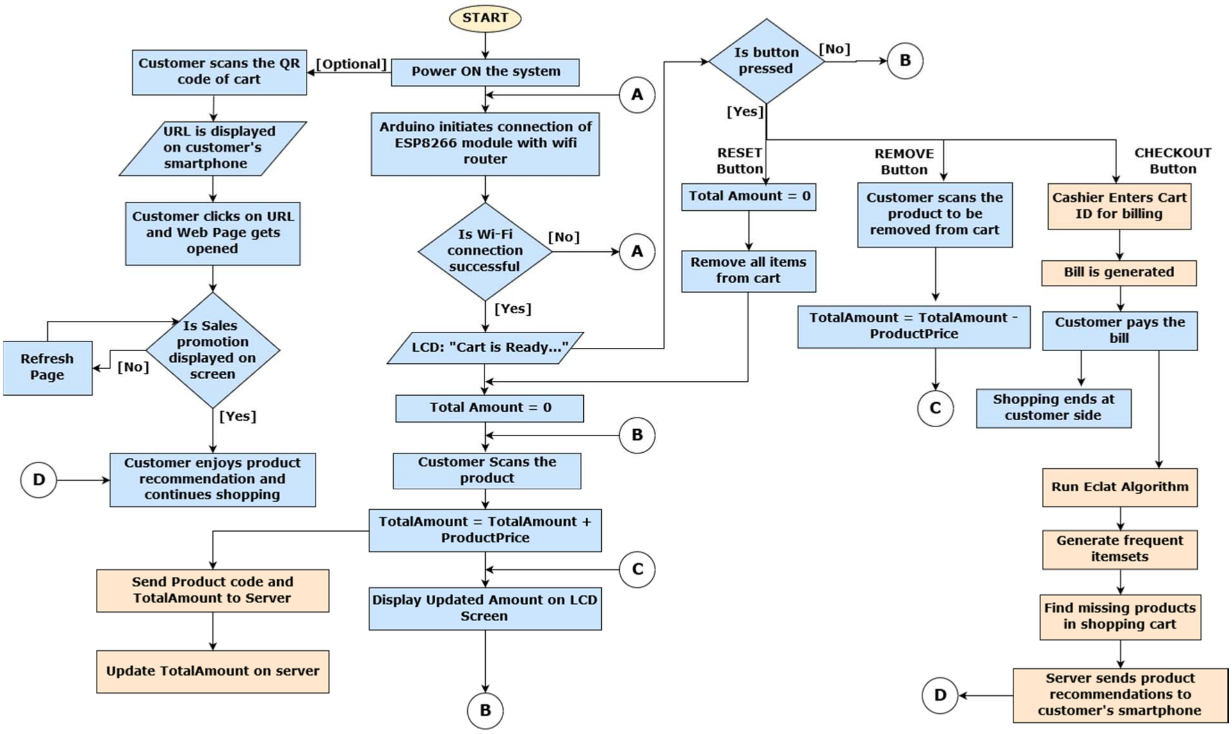

Fig. 4. The flow of SCAR system

V. EXPERIMENTATION

For implementing this system, hardware used is Arduino ATmega2560, ESP8266 Wi-Fi module, 16x2 LCD Display, MIRF522 RFID Reader, TTP-223 Touch keys, buzzer, jumper wires, battery, and the capacitor. MIRF522 RFID Tags are required to be attached to the product for scanning the product. The required software setups are JRE 1.8, XAMPP server with MySQL, Apache, Tomcat, and Arduino IDE, Python 3.6.

(a) Initial message on screen.
(b) Cart ready message.
(c) Product adding message.
(d) Displays updated bill.
(e) When REMOVE button is pressed.
(f) Product removed message.
(g) Cart RESET confirmation prompt.
(h) Cart RESET done.

Fig. 5. Messages displayed on the cart's 16x2 LCD Display.

Fig. 5 show the event messages displayed on cart's 16 x 2 display screen. The message of fig. 5 (a) is displayed at Powering ON of the SCAR system. Once the cart is connected to supermarket's Wi-Fi access point, it displays the cart ready message as shown in fig. 5 (b). When you scan the product for adding it to the cart, product adding message is displayed as shown in fig. 5 (c). At every scan, the updated bill is displayed on the screen, for e.g. fig. 5 (d). On pressing the REMOVE button, a message prompts on screen for scanning the product to be removed (fig. 5 (e)). After scanning the product to be removed it displays the "Item removed" message on the screen (fig. 5 (f)). To quit the shopping or making the cart empty, RESET button is required to press, which prompts a confirmation message on the screen as shown in fig. 5 (g). On pressing the OK button within 5 seconds, cart gets reset and displays the message on the screen shown in fig. 5 (h).

Customers can check their product list by scanning the QR code provided on the cart, through which they can also view the product recommendations sent by server related to their products present in the cart. Fig 6.1 shows the top-selling products in real time, which are generated by executing the CPR algorithm. Also, if the customer is having more than three products in their cart, then they will get the recommendation of different types of carrying bags. Fig 6.2 shows the recommended products those are related to their cart's product which is displayed in fig 6.4. Remaining frequently bought products, those are not related to their cart's products are displayed as frequently sold products. Also, the customer can use the app mode, which is similar to online shopping. After checkout, they can get their products in hand before leaving the supermarket. This will save their time for searching products and also with recommendations the location of the product's rack is also provided.

Fig. 6. Product Recommendation on the customer's smartphone

VI. RESULTS AND DISCUSSION

For experimentation, we used the latest available version HPE LoadRunner 12.55 tool which is developed by Micro Focus. This tool is used for testing software applications, measuring the behavior of the system, and mainly, performance under load by simulating 50 number of users simultaneously. By using the Virtual User Generator tool of HPE LoadRunner, we recorded script. In a controller application of LoadRunner we designed a scenario to simulate real-time SCAR system behavior. Later, we executed the designed scenario for 8 hours using 50 virtual users. On completion of the test we used an analysis tool of LoadRunner to evaluate the performance of designed SCAR system. The system's performance is based on the response time of recorded transaction in the script. The performance of the system is analyzed in the form of various graphs.

Table II and III provide the analysis and statistics summary, which gives period and the duration of the analysis with the statistics of SCAR system behavior. Table III shows the transaction summary of the simulated scenario. In this table, all details related to the recorded transaction, such as, average transactions of actions performed by employees and customers while shopping, as well as browsing for recommended products is tabulated.

TABLE II. ANALYSIS & STATISTICS SUMMARY

Period: 10-May-18 12:40:29 AM to 18 9:08:21 AM (India Standard Time)	
Duration: 8 hours, 27 minutes and 52 seconds.	
Maximum Running Vusers:	50
Total Throughput (bytes):	21,497,471,028
Average Throughput (bytes/second):	705,460
Total Hits:	991,202
Average Hits per Second:	32.527

TABLE III. TRANSACTION SUMMARY

Transaction Name	Minimum	Average	Maximum	90 Percent	Pass	Fail
Customer_1_Page_Load	0.088	0.122	29.722	0.137	5,715	0
Customer_2_Sign_In	0.03	0.035	0.295	0.034	5,715	0
Customer_3_Select_Category	0.004	0.076	0.601	0.164	12,667	0
Customer_4_Add_To_Cart	0.04	0.128	1.555	0.213	12,667	0
Customer_5_Continue_Shopping	0.002	0.003	0.104	0.003	12,667	0
EMP_1_Page_Load	0.002	0.003	0.096	0.004	5,715	0
EMP_2_Sign_In	0.005	0.013	0.23	0.021	5,715	0
EMP_3_View_Cart	0.003	0.008	0.225	0.009	5,715	0
EMP_4_Make_Payment	2.125	3.189	17.223	5.425	5,715	0
EMP_5_Sign_Out	0.002	0.006	0.129	0.006	5,715	0
Sales_Promotion_1_Page_Load	0.009	0.067	0.796	0.12	12,667	0
Sales_Promotion_2_Click_Cart	0	0	0.001	0	12,667	0
Sales_Promotion_3_Click_Back	0.009	0.037	0.377	0.059	12,667	0

Following graphs are generated using the LoadRunner analysis tool after execution of designed scenario.

Fig. 7 shows the graph of virtual running users concurrently during the whole scenario execution. Which indicates the maximum running users are 50 (The free version of HPE's LoadRunner Community edition offers the maximum 50 number of virtual users).

Fig. 8 shows the graph of hits per second by virtual users, which indicates maximum hits are reached to approximately 35 hits per second to SCAR system.

Fig. 9 shows the average transaction response time for each action performed by customers and employees.

Fig. 10 shows the throughput of the SCAR system in bytes per second. The average throughput of the system is 705,460 bytes per second.

Fig. 11 shows the number of transactions for each action performed by customers for adding products, for browsing products through recommendations, viewing details of products dropped in the cart, and actions performed by the employee for billing and making payments.

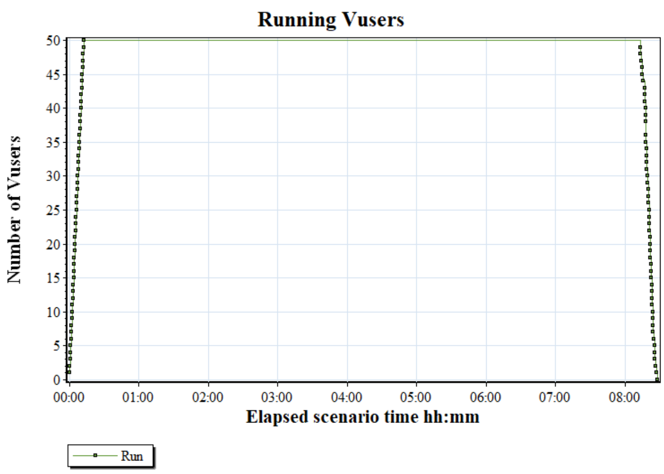

Fig. 7. Graph of running user (Vusers/virtual users) concurrently

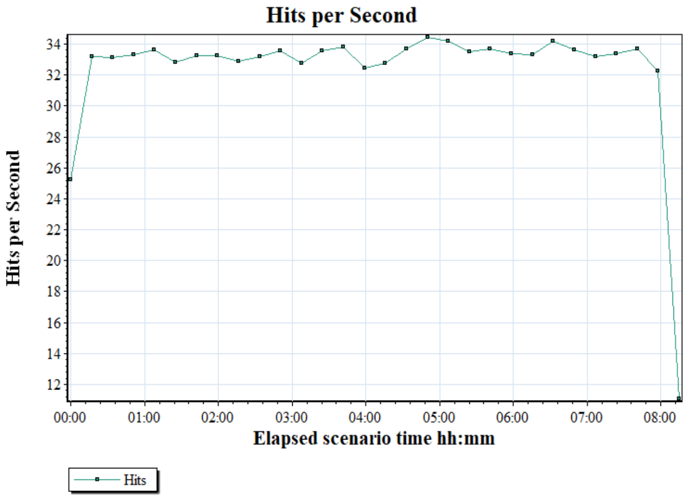

Fig. 8. Users' hits per second

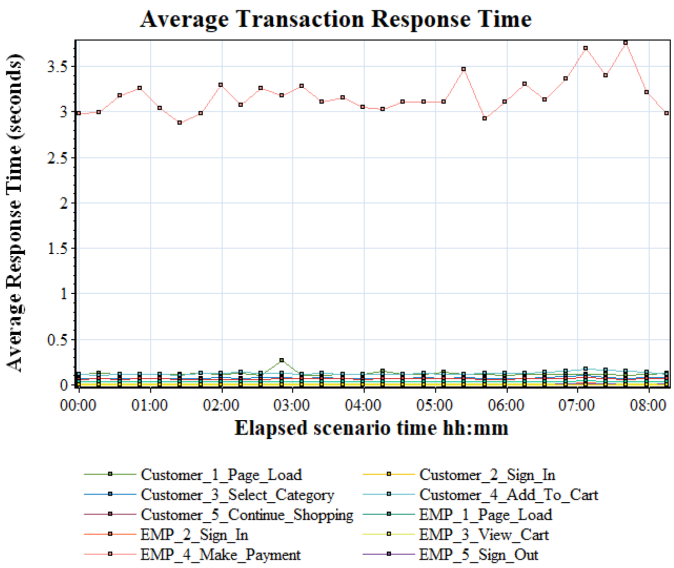

Fig. 9. Conceptual view of SCAR system.

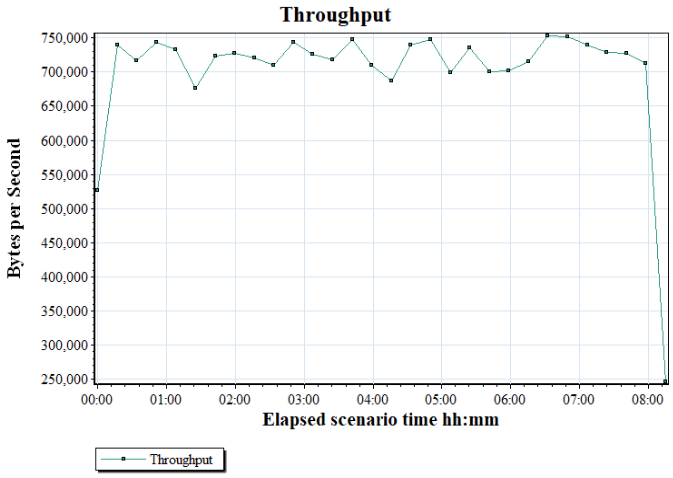

Fig. 10. Throughout of SCAR system.

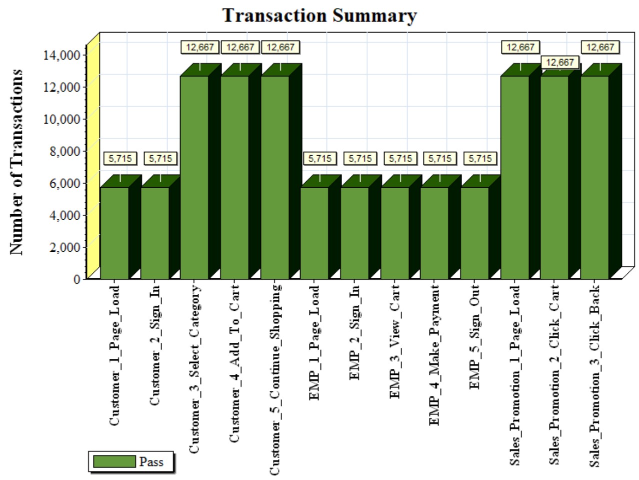

Fig. 11. Transaction summary.

VII. Conclusion

Product recommendation in online shopping websites is normally achieved by association rule mining algorithms. This system will be the first offline retail system, that will recommend products to customers based on the products dropped in the cart by customers. In this SCAR system, combining ARM algorithm with RFID technology for sales marketing and quick billing service will give the customers ease of shopping and also increases the sales of products. The implementation of this SCAR system in real time is feasible if barcode system is replaced by the RFID system at product manufacturing level, as it is not feasible to attach RFID tags to each product manually.

VIII. Future Work

SCAR system can be extended to smart malls by adding Long-Range RFID readers and Artificial Intelligence for the stock control system to place orders automatically. Following are some features that can be extended:

- Installing Long-Range RFID readers at entry gates for stock entry to update the database automatically.
- Implementing a stock control system using Long-Range RFID Readers and Artificial Intelligence.
- Alerting for misplaced products to place them correctly.

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References

[1] P. Rani and I. Ranjan, “A comprehensive study of map-reduce frequent itemset mining: a survey”, International Journal of Control Theory and Applications, Vol. 9 Issue 46, pp.31-44, 2016.

[2] A. Saxena, and S. Gadhiya, “A survey on frequent pattern mining methods apriori, eclat, FP growth”, International Journal of Engineering Development and Research, Vol. 2, Issue 1, pp. 92-96, 2014.

[3] P. Shah, J. Jha, N. Khetra, and M. Zala, “A literature review on improving error accuracy and range based on RFID for smart shopping”, IJSRD - International Journal for Scientific Research & Development, Vol. 3, Issue 10, 2015, ISSN (online): 2321-0613.

[4] A. Yewatkar, F. Inamdar, R. Singh, Ayushya, A. Bandal, “Smart cart with automatic billing, product information, product recommendation using RFID & ZigBee with anti-theft”, 7th International Conference on Communication, Computing, and Virtualization 2016.

[5] Y. Chen, Z. Zheng, S. Chen, L. Sun, and D. Chen. "Mining Customer Preference in Physical Stores from Interaction Behavior." IEEE Access, 5, pp.17436-17449.

[6] Y. Wang, and C. Yang, “3S-cart: A lightweight, interactive sensor-based cart for smart shopping in supermarkets”, IEEE Sensors Journal, Vol. 16, No. 17, September 1, 2016.

[7] Yao, Jiawei, Jiajun Yao, Rui Yang, and Zhenyu Chen. "Product recommendation based on search keywords." In Web Information Systems and Applications Conference (WISA), 2012 Ninth, pp. 67-70. IEEE, 2012.

[8] B. Goethals, “Survey on frequent pattern mining,” University of Helsinki, 2003.

[9] A. Mahanti, and R. Alhajj, “Visual interface for online watching of frequent itemset generation in apriori and eclat”, Proceedings of the Fourth International Conference on Machine Learning and Applications (ICMLA’05) IEEE, 2005.

[10] J. Heaton, "Comparing dataset characteristics that favor the Apriori, Eclat or FP-Growth frequent itemset mining algorithms." In *SoutheastCon, 2016*, pp. 1-7. IEEE, 2016.